

ALGEBRA GENERAL EXAM, JANUARY 6, 2013, 9AM-1PM

Directions.

- Show all your work and justify any statements that you make.
- State clearly and fully any theorem you use.
- Vague statements and hand-waving arguments will hurt your grade.
- You may assume the statement in an earlier part proven in order to do a later part.
- Do each problem on a separate one-sided sheet of paper, and staple them together in the correct order.

Problem 1

(10 points). Let K be an algebraically closed field. Show that any element of finite order in $GL_n(K)$ is diagonalizable. (Hint: Jordan Form!).

Problem 2

(10 points). Find all ring homomorphisms

(1) from $\mathbb{Z}$ to $\mathbb{Z} / 30\mathbb{Z}$;

(2) from $\mathbb{Z} / 30\mathbb{Z}$ to $\mathbb{Z}$.

Problem 3

(10 points). Let $\mathbb{Q}(\sqrt{-2})$ be a quadratic field with associated ring of integer $\mathcal{O} = \mathbb{Z}[\sqrt{-2}]$. Prove that $\mathcal{O}$ is a Euclidean Domain . (Hint: use the field norm.)

Problem 4

(10 points). Prove that if R is a principle ideal domain (P.I.D.) and D is a multiplicatively closed subset of R with $0 \notin D$, then $D^{-1}R$ is also a P.I.D.

Problem 5

(10 points).

(1) Consider the abelian group

$$A = \prod_{n \geq 2} \mathbb{Z}/n\mathbb{Z}$$

Show that this is not a torsion group by exhibiting an element of infinite order.

(2) Show that $\mathbb{Q} \otimes_{\mathbb{Z}} A \neq 0$. (Hint: this is $S^{-1}A$ for $S = \mathbb{Z} - \{0\}$.) Bonus: Determine $\dim_{\mathbb{Q}}(\mathbb{Q} \otimes_{\mathbb{Z}} A)$.

Problem 6

(10 points). Let $K \mid F$ be an extension of finite fields, and let L, M be subfields of K containing F . Assume that $L \cap M = F$.

(1) Show that the degrees $[L : F]$ and $[M : F]$ are relatively prime. Hint: How many subfields of a given order does a finite field have?

(2) Now assume additionally that $L = F(\alpha)$, $M = F(\beta)$ and $K = F(\alpha, \beta)$ for some $\alpha, \beta \in K$. Prove that $K = F(\alpha + \beta)$.

Problem 7

(10 points). Consider the real number $u = \sqrt{3 + \sqrt{11}}$.

(1) Determine the minimal polynomial for u over $\mathbb{Q}$, and justify that it is the minimal polynomial.

(2) Is $\mathbb{Q}[u]$ the splitting field for the minimal polynomial of u ? (Hint: consider which roots are real and which are complex.)

(3) Determine the Galois group of the splitting field. (Hint: What is the degree of the field extension?)

Problem 8

(10 points). Use the semidirect product constructions to classify the groups of order 44. (Hint: start by analyzing Sylow subgroup structures.)